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A Robust Path Tracking Controller for Autonomous Mobility with Control Delay Compensation Using Backstepping Control

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Abstract: Control delay phenomena, such as time delays and actuator lags, can compromise the control performance of autonomous mobility systems, leading to increased control errors. Therefore, it is essential to develop a control delay compensation algorithm. This paper proposes a Lyapunov-based backstepping steering control algorithm to compensate for control delays in autonomous mobility systems. To estimate the control delay in the steering system, the Recursive Least Squares (RLS) algorithm was employed to calculate the time constant in real time. The control delay was estimated using an RLS designed based on a first-order differential equation. A backstepping steering controller was developed to calculate the desired steering angle using simplified error dynamics for reference path tracking. The control errors, specifically the lateral preview and yaw angle errors, were derived by calculating the path error between the current position and the waypoint. The performance of the proposed control algorithm was evaluated using the DC motor and CarMaker software 8.1.1(IPG Automotive, Karlsruhe, Germany) under scenarios involving sinusoidal input and four-curved loop and S-curved paths respectively.

Keywords: control delay compensation; lyapunov stability; backstepping control; recursive least squares; autonomous mobility

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1. Introduction

In recent decades, advancements in electrical and electronic components, battery technologies, and the development and structure of big data systems have become feasible. Additionally, an increase in computational power has enabled the application of advanced algorithms that consider the driving stability and energy efficiency of hybrid and autonomous vehicles [1,2]. Accordingly, the commercialization of autonomous mobility systems is accelerating in various areas, and mobility solutions capable of performing multiple functions are being developed across various industries using these systems [3,4]. In these systems, a significant number of sensors, actuators, and electronic control units (ECUs) are used, which can lead to control delays caused by hardware (such as physical loads) [5,6] and software (including computational loads and communication delays) [7,8]. When transmitting control signals over a network, data loss or signal delays may occur due to network traffic or limitation of network bandwidth. In these systems, the time delay within the control loop presents a significant challenge that can lead to stability issues. Additionally, these delays reduce the response speed of the controller, making it challenging to maintain the desired performance. This can result in system instability or quantization errors, which complicate real-time control [9,10]. Control delays in an over-actuated system, which include both delays and actuator lags, can negatively affect the control performance, stability, and operator safety. Therefore, to ensure optimal performance, it is essential to develop a control methodology that compensates for these delays. Additionally, when designing a control algorithm based on a mathematical model to ensure

control robustness, a precise understanding of the system parameters can significantly influence control performance.

Studies are being conducted to estimate and compensate for delays in networks [11,12], software [13,14], and actuators [15,16] across various systems. These studies aim to enhance the stability of the systems and target performance by applying mathematical model-based control, optimal control, and adaptive theory. Therefore, we propose a methodology that estimates control delay as a system parameter in real time and considers it when deriving the steering control input for autonomous mobility. The main contributions of this study are as follows:

1. Develop a backstepping-based steering control algorithm for robust path tracking that satisfies Lyapunov stability conditions while considering control delays, through a straightforward computational process based on system dynamics.
2. Design of real-time control delay estimation algorithm using first-order differential equation of the steering system, using RLS with forgetting factor.
3. Validation of real-time control delay estimation algorithm on DC motor platform.

Ackermann kinematics-based simplified error dynamics were utilized to design a backstepping-based steering control algorithm to track the reference path. Control errors were derived based on the waypoints of the target path, considering the lateral preview and yaw angle errors. The steering angle, compensating for the control delay, was calculated using the system dynamics expressed by a first-order differential equation that included the control delay. The control delay was applied to the value estimated in real time using the RLS. The remainder of this paper is organized as follows:

Section 2 discusses related work, and Section 3 describes the design process of the proposed steering control algorithm. Section 4 presents a performance evaluation based on simulations, and Section 5 discusses the conclusions and future work.

2. Related Works

Intelligent programs and robots can enhance production because they only require electrical energy while humans need rest and food. For use in industrial fields such as factories, medical institutions, and farms, various types of mobility are being developed to suit specific purposes, allowing workers to operate more efficiently [17]. In factories, mobile robots can coexist with humans and navigate throughout the factory to assist in storage, transportation, and production processes. They utilize LiDAR and cameras to identify current and target locations based on precise maps, and execute the corresponding location or task through actuator control [18,19]. In hospitals, Internet of Things (IoT) modules are utilized to monitor and control autonomous mobility for transporting medicine and food to patients as well as to perform disinfection tasks within the facility [20,21]. In addition, farms use autonomous mobility to identify target crops, assess growth data, and facilitate harvesting. Based on various sensors and actuators, it detects target crops and obstacles, minimizing crop damage, and performs path planning to reach the target quickly and accurately to achieve a specific goal [22–24].

Robust control algorithms are essential for the effective and consistent execution of various types of mobility robots. Park et al. proposed a Linear Quadratic Regulator (LQR) method to enhance the safety of autonomous vehicles by executing drift maneuvers that induce rear-wheel slip under extreme conditions, such as on slippery roads. They calculated the drift equilibrium state using nonlinear vehicle dynamics and tire models, and maintained this equilibrium by simultaneously controlling the steering torque and rear-wheel torque based on the LQR [25]. Amertet et al. developed a kinematic model for a four-wheel skid-steering mobile robot and proposed an effective control methodology to address changes in the dynamics of the model caused by defects, uncertainties, and parameter variations. This was achieved by calculating the Q and R matrices of the LQR based on observability [26].

The over-actuated system must consider constraints on both the actuators and state variables. Accordingly, Model Predictive Control (MPC) is utilized, offering the advantage

of deriving realistic optimal control inputs by considering the constraints [27,28]. Hadian and Zhang proposed an algorithm that predicts the robot's state using a recurrent neural network and incorporates this prediction into robust path tracking control based on MPC. They expressed the complex nonlinear system of a mobile robot as a quasi-linear parameter-varying model and approximated it with a linear model [29]. Additionally, it is utilized as a longitudinal and lateral control algorithm based on Sliding Mode Control (SMC), which is robust control methodology against external disturbances and system uncertainties [30,31]. Thus, it can be confirmed that control methodologies such as MPC, SMC, and LQR exist to secure reasonable control performance of the system. Furthermore, with the advancement of Artificial Intelligence (AI), Deep Learning techniques are increasingly being employed the self-tuned variables of control algorithms [32,33] or to derive control inputs [34,35].

To ensure and enhance the stable performance of the mathematical models and various AI-based control algorithms, it is essential that they rely on accurate system parameters and state variables. Accordingly, the uncertainty in the observer-based system and external disturbances [36–38] were addressed using the vehicle's internal sensors and estimation algorithms. These algorithms incorporate parameters such as the vehicle's center point, tire stiffness [39], and road friction coefficient [40], utilizing unscented, extended, and adaptive Kalman filters. Additionally, factors such as the body slip angle [41,42] and longitudinal velocity [43] were considered.

Autonomous mobility systems require additional sensors, actuators, and control devices. Therefore, random network delays may arise due to limitations in vehicle network bandwidth, computational load, and external disturbances. This may result in excessive performance degradation and instability within the system; therefore, an algorithm that estimates and compensates for control delay is necessary. Research has been conducted to estimate network delays using machine learning techniques [44] and mathematical modeling [45]. Additionally, efforts are underway to estimate Vehicle-to-Vehicle (V2V) communication delay [46] and incorporate this information into the controller for compensation. Furthermore, neglecting the steering delay, which is the time required to generate the tire force, can increase the model's uncertainty and degrade performance [47]. Consequently, research has focused on estimating and compensating for actuator delays in conjunction with network delays [48–51].

Through related research, various strategies to estimate and compensate for control delays have been proposed, but real time estimation and compensation algorithms have been insufficiently performed. Section 3 describes the design process for the proposed steering control algorithm.

3. Control Delay Compensated Path Tracking Control

Control delays, such as actuator lags and communication delays, can increase control errors and decrease control performance. In this study, to address the control performance degradation caused by the control delay of the steering system, a Lyapunov-based backstepping steering control algorithm is used to compensate for the control delay in the steering system by estimating the control delay in real time. A backstepping-based steering controller was designed to compensate for the control delay based on the Lyapunov stability condition. To estimate the control delay in real time, an RLS with a forgetting factor was applied. Figure 1 illustrates a block diagram of the proposed algorithm, which comprises backstepping-based steering control and RLS-based control-delay estimation for control delay compensation.

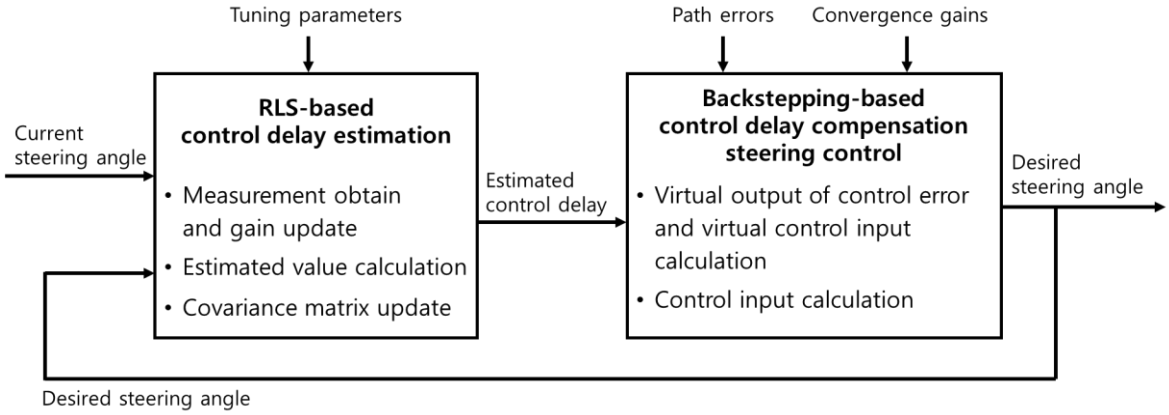


Figure 1. Block diagram of the proposed algorithm.

3.1. Lyapunov Stability-Based Backstepping Steering Controller

In this study, a steering control algorithm for tracking the reference path was deduced using simplified error dynamics based on Ackerman kinematics, and the control input was calculated using the Lyapunov approach-based backstepping control. To derive simplified error dynamics, a bicycle model based on Ackerman kinematics was applied, considering 3 DOF planar motion, as shown in Figure 2.

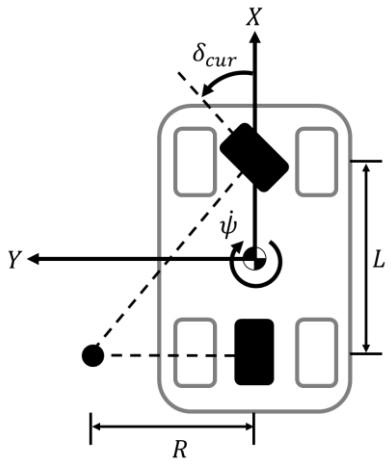


Figure 2. Ackerman kinematics-based bicycle model.

According to Ackerman kinematics, the longitudinal velocity is defined as Equation (1), assuming that the vehicle satisfies uniform circular motion.

$$v_x = R\dot{\psi} \tag{1}$$

where v_x represents the longitudinal velocity, R represents the radius, and $\dot{\psi}$ denotes the yaw rate respectively. The radius can be derived as follows:

$$R = \frac{L}{\delta} \tag{2}$$

where L is the wheel base and δ is the steering angle, respectively. The yaw rate can be derived using the longitudinal velocity, wheel base and steering angle, as shown in Equation (3), under the assumption of a small steering angle.

$$\dot{\psi} = \frac{v_x}{L}\delta \tag{3}$$

$$where\ tan(\delta) \approx \delta$$

The error dynamics for tracking the reference path were designed based on Ackerman kinematics-based simplified error dynamics, as follows:

$$\dot{e}_\psi = \dot{\psi}_{cur} - \dot{\psi}_{des} \tag{4}$$

$$\dot{e}_y = v_x e_\psi \tag{5}$$

where e_ψ and e_y denote the yaw angle error and lateral error respectively, and $\dot{\psi}_{cur}$ is the current yaw rate, $\dot{\psi}_{des}$ is the desired yaw rate, v_x denotes the longitudinal velocity.

To converge the lateral error to zero, the relationship between the lateral error and virtual control input for lateral error u_1 is defined by Equation (6). If the virtual control input for lateral error is designed as Equation (7), the lateral error satisfies the Lyapunov stability condition.

$$\dot{e}_y = v_x u_1 \tag{6}$$

$$u_1 = -\frac{\eta_1}{v_x} e_y\ (where,\ \eta_1 > 0) \tag{7}$$

where η_1 denotes the convergence rate of the lateral error. It is designed as a tuning parameter and determines reasonable value through evaluation.

Then, the virtual control output of the yaw angle error y_1 can be defined using the virtual control input for the lateral error u_1, as shown in Equation (8). By Equations (1), (2) and (8), the time-differential equation of the virtual control output for the yaw angle error can be rewritten as Equation (9).

$$y_1 = e_\psi - u_1 \tag{8}$$

$$\dot{y}_1 = \dot{e}_\psi - \dot{u}_1 = \frac{v_x}{L}\delta_{cur} - \dot{\psi}_{des} - \dot{u}_1 \tag{9}$$

where δ_{cur} denotes the current steering angle.

Then, the current steering angle can be derived using the convergence rate of the lateral error and the virtual control output of the lateral error, as in Equation (10).

$$\delta_{cur} = -\frac{L}{v_x}\alpha_1\eta_1 - \frac{L}{v_x}(\eta_1 + \alpha_1)e_\psi + \frac{L}{v_x}\dot{\psi}_{des} \tag{10}$$

$$where, \dot{y}_1 = -\alpha_1 y_1\ (\alpha_1 > 0)$$

where α_1 denotes the convergence rate of output of lateral error.

To derive the virtual control input for steering angle u_2, the time differential of the yaw angle error can be formulated as follows:

$$\dot{e}_\psi = \frac{v_x}{L}u_2 - \dot{\psi}_d \tag{11}$$

To converge the yaw angle error to zero, the convergence rate of the yaw angle error was defined to satisfy the Lyapunov stability condition.

$$\dot{e}_\psi = -\eta_2 e_\psi\ (where, \eta_2 > 0) \tag{12}$$

where η_2 represents the convergence rate of the yaw angle error and it is set at a reasonable value through the experiment. The virtual control input for the steering angle can be derived using Equation (13), by subtracting Equations (11) and (12), respectively.

$$u_2 = \frac{L}{v_x}\left(-\eta_2 e_\psi + \dot{\psi}_d\right) \tag{13}$$

To converge the virtual control input for the steering angle to the current steering angle, the output of the lateral error is defined as

$$y_2 = \delta_{cur} - u_2 \tag{14}$$

The first-order differential equation of the steering system can be expressed using the time constant in Equation (15).

$$\dot{\delta}_{cur} = \frac{1}{\tau_d}(\delta_{des} - \delta_{cur}) \tag{15}$$

where τ_d denotes the time constant of steering system. By subtracting Equations (14) and (15), Equation (16) can be derived.

$$\dot{y}_2 = \frac{1}{\tau_d}\delta_{des} - \frac{1}{\tau_d}\delta_{cur} - \dot{u}_2 \tag{16}$$

Using the Lyapunov stability condition, the time differential of the virtual output of the lateral error is defined in Equation (17) with the convergence rate.

$$\dot{y}_2 = -\alpha_2 y_2 \tag{17}$$

The desired steering angle can then be expressed using Equation (18) by Equation (15). The time differential of the virtual output of the lateral error can be expressed as shown in Equation (19).

$$\delta_{des} = \delta_{cur} + \tau_d \dot{y}_2 + \tau_d \dot{u}_2 \tag{18}$$

$$\dot{y}_2 = \left(\eta_2 - \frac{1}{\tau_d}\right)\delta_{cur} + \frac{1}{\tau_d}\delta_{des} - \frac{\eta_2}{v_x}L\dot{\psi}_{des} - \frac{L}{v_x}\ddot{\psi}_{des} \tag{19}$$

Finally, the desired steering angle is derived using Equation (20), by subtracting Equations (9), (18) and (19), respectively.

$$\delta_{des} = (1 - \tau_d(\alpha_2 + \eta_2))M - N \tag{20}$$

$$where,\ M = -\frac{L}{v_x}\left(\alpha_1\eta_1 e_y + (\alpha_1 + \eta_1)e_\psi\right),\ N = \frac{\tau_d L}{v_x}\alpha_2\eta_2 e_\psi$$

3.2. RLS-Based Control Delay Estimation

The control delay of an autonomous mobility steering system can vary not only physically but also computationally in real time. Therefore, it is necessary to develop a real time control delay estimation algorithm.

In this study, to estimate the control delay of the steering system of autonomous mobility in real time, an RLS with the forgetting factor methodology has been designed using the first-order differential equation of the steering system, as shown in Equation (15).

$$\delta_d - \delta_c = \dot{\delta}_c \hat{\tau}_d \tag{21}$$

where $\hat{\tau}_d$ is the estimated time constant. To calculate the control delay, the difference in the desired and current steering angles is considered as the output of the linear system (y), and the time differential of the current steering angle is considered as the regressor of the RLS (ϕ).

$$y = \phi\hat{\theta} \tag{22}$$

where $\hat{\theta}$ represents the estimated value. In this study, $\hat{\theta}$ is defined the estimated control delay of steering system. The defined cost function is used to derive the estimated value by considering the previous data using forgetting factors.

$$J = \sum_{i=0}^{k-1} \lambda^{(k-1)-i} \left| y_i - \phi_i \hat{\theta}_i \right|^2 \tag{23}$$

where λ denotes the forgetting factor and it is determined using the trial and error method. The estimated value is calculated by updating the updating gain (L), the estimated value ($\hat{\theta}$), and the covariance matrix (P) in discrete time with Equations (24)–(26).

$$L_k = P_k \phi_k \left(\lambda + \phi_k^T P_k \phi_k \right)^{-1} \tag{24}$$

$$\hat{\theta}_k = \hat{\theta}_k + L_k \left(y_k - \phi_k \hat{\theta}_k \right) \tag{25}$$

$$P_k = \left(I - L_k \phi_k^T \right) P_k \frac{1}{\lambda_k} \tag{26}$$

4. Performance Evaluation

To validate the proposed algorithms, Evaluations 1 and 2 were performed using MATLAB/Simulink, actual DC motor, and CarMaker software 8.1.1 respectively. The RLS-based control delay estimation algorithm was evaluated using the MATLAB/Simulink and actual DC motor platform in real-time in Evaluation 1. Figures 3 and 4 show the used DC motor platform and model schematic for evaluation, respectively.



Figure 3. DC motor platform.

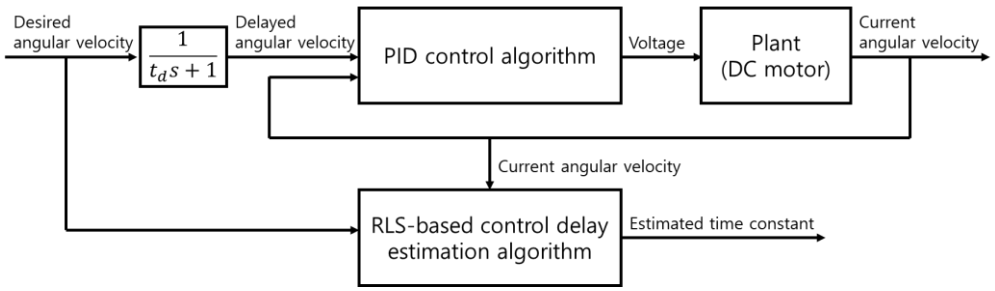


Figure 4. Architecture of the performance evaluation.

The PID controller was applied to track the desired delayed angular velocity by calculating the voltage input. The control delay was applied using the first-order differential

equation function in Equation (15) in a Simulink environment. The RLS-based control delay estimation algorithm was used to estimation the control delay of the DC motor in real-time. Table 1 describes the DC motor specification and parameters used for evaluation.

Table 1. Parameters for evaluation 1.

Division		Value
PID controller	P gain	0.15
	I gain	2
	D gain	0.001
RLS	Initial value of covariance matrix	100,000
	Initial value of estimated value	1
	Forgetting factor	0.995

Then, at Evaluation 2, the integrated algorithm with control delay estimation and steering control was evaluated using MATLAB/Simulink and CarMaker software by constructing a waypoint-based path error calculation algorithm and a virtual driving environment. The waypoint-based path error calculation algorithm was designed to calculate the lateral, lateral preview, and yaw angle errors at the preview point. Figure 5 illustrates the architecture of the performance evaluation environment.

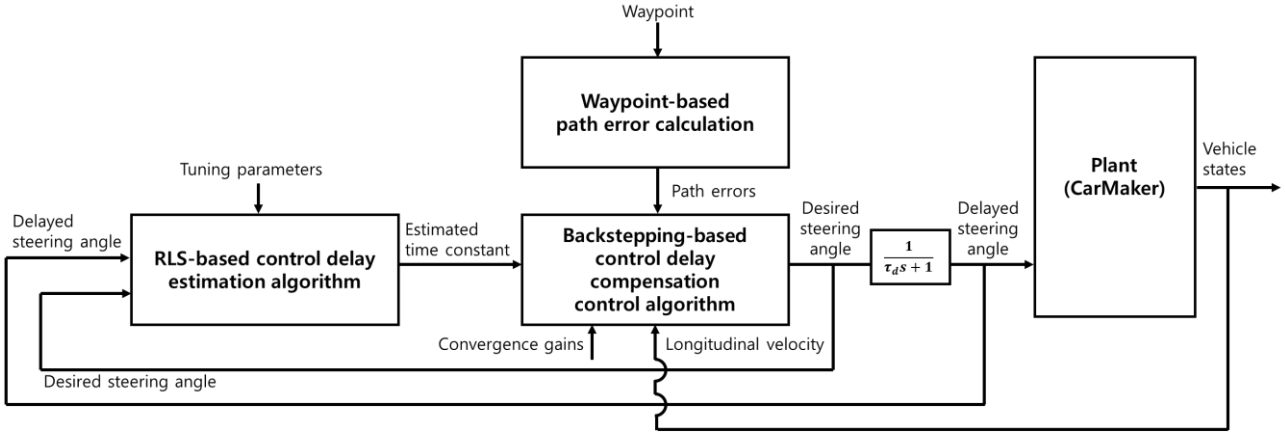


Figure 5. Architecture of the performance evaluation.

In the waypoint-based path error calculation block, path errors, including the lateral error, lateral preview error, and yaw angle error, were calculated based on the current position of the ego vehicle and predefined waypoints. The evaluation scenarios adopted an S-curved path and a four-curved loop path. The longitudinal velocity was maintained constant at 30 km/h and 60 km/h in CarMaker, and the control delay was set to 0.2 s [51] in Scenarios 1 and 2. The control delay is applied by designing the first-order differential equation function in Simulink as shown in Equation (15). The evaluation results were analyzed by comparing the proposed algorithm with a backstepping controller without control delay compensation and with SMC. Table 2 describes the control parameters for Evaluation 2.

Table 2. Control parameters for Evaluation 2 (Scenarios 1, 2).

Division		Value
Backstepping controller	Convergence rate of lateral error	10
	Convergence rate of yaw error	0.5
	Convergence rate of lateral output	10
	Convergence rate of yaw output	0.5
RLS	Initial value of covariance matrix	0.01
	Initial value of estimated value	0.01
	Forgetting factor	0.9993
SMC	Weighting of lateral error	5
	Disturbance boundary	10
	Decay rate of the Lyapunov function	0.5

The SMC was designed to track paths, which are yaw angle error and path error, to zero by defining the sliding surface as shown in Equation (27).

$$\sigma = e_\psi + Ce_y \tag{27}$$

where σ is the sliding surface and C is the weighting of the lateral error. C is set using trial and error method to reasonable result. The cost function was defined based on the Lyapunov function in Equation (28).

$$J = \frac{1}{2}\sigma^2 \tag{28}$$

Then, the control input is derived to satisfy the Lyapunov stability condition.

$$\delta_{d,SMC} = -\left(L_b + \frac{\alpha}{\sqrt{2}}\right) sign(\sigma) \tag{29}$$

where L_b denotes the disturbance boundary and α represents the decay rate of the Lyapunov function. The results of the evaluation are analyzed as follows [52].

4.1. Evaluation 1 (Using the DC Motor)

The results of Evaluation 1, which is the control delay estimation algorithm using the RLS with forgetting factor validation, are analyzed through Figures 6–8.

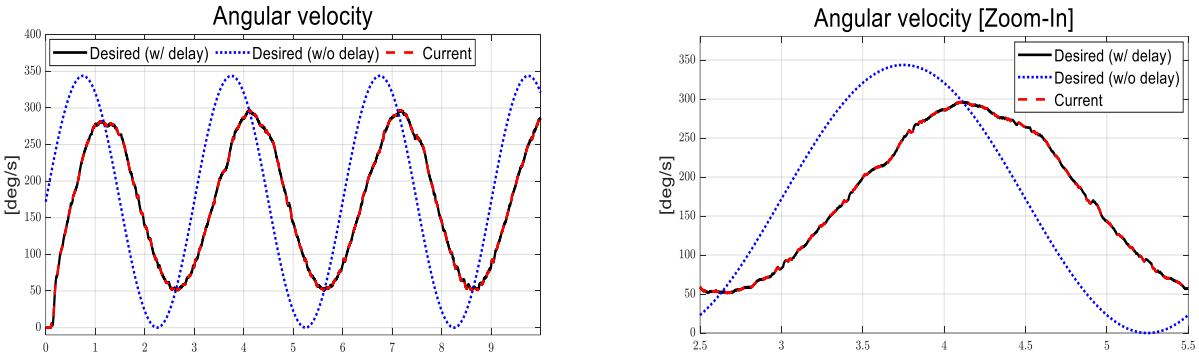


Figure 6. Evaluation 1: angular velocity.

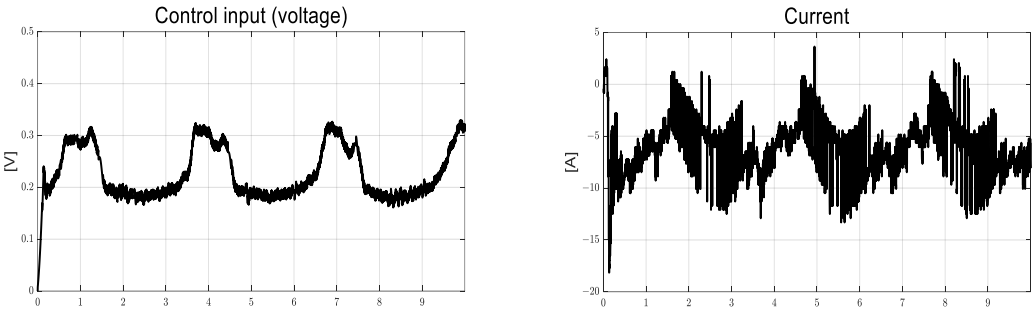


Figure 7. Evaluation 1: control input and current.

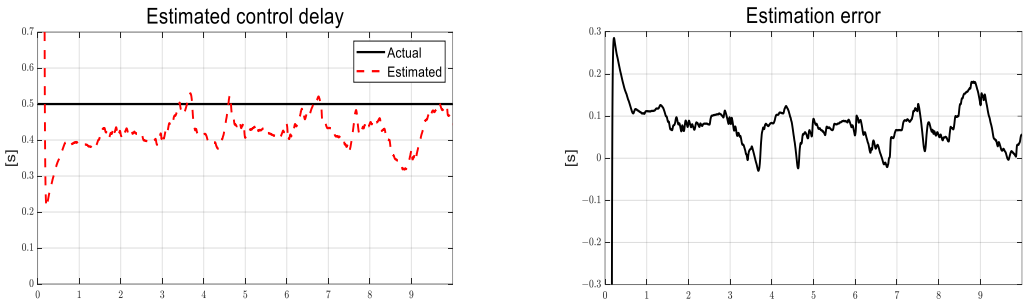


Figure 8. Evaluation 1: estimated control delay and estimation error.

Figure 6 shows the desired and current angular velocities, respectively. The desired angular velocity was applied with the control delay using time constant 0.5 s. Figure 7 shows the current and voltage as control inputs, with the voltage being regulated by a PID controller real-time. As shown in Figure 7, the delayed desired voltage can be tracked reasonably using the PID controller. Figure 8 shows the actual and estimated control delay and estimation error. Although there was an estimation error at the initiation of the evaluation, it converged within 0.5 s. The average estimation error after 0.5 s was approximately 0.065 s. The rate of change of the error was calculated using the linear Kalman filter, and it was confirmed that the estimation error occurred relatively. In the future, when applying the algorithm to a real platform, it is essential to consider uncertainties and disturbances.

4.2. Evaluation 2—Scenario 1 (S-Curved Path)

The backstepping steering control algorithms—with and without control delay compensation—and SMC were compared based on the results of Evaluation 2. Figures 9–15 display the results of the performance evaluation for Evaluation 2—Scenario 1.

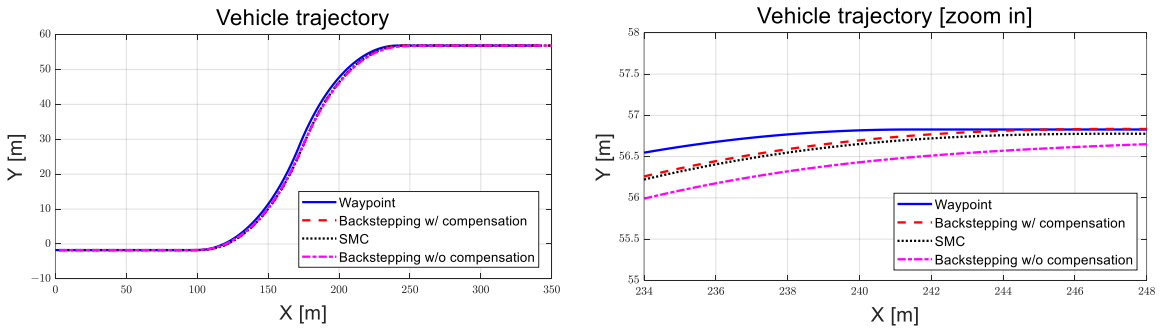


Figure 9. Scenario 1: vehicle trajectory.

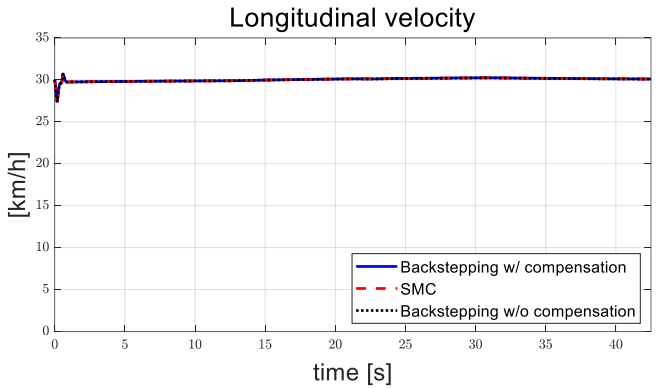


Figure 10. Scenario 1: vehicle longitudinal velocity.

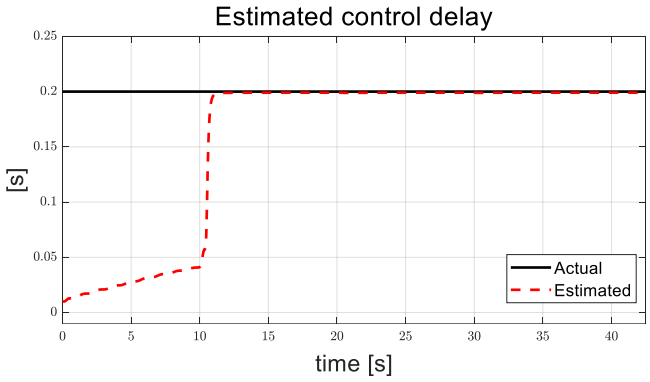


Figure 11. Scenario 1: estimated control delay of the steering system.

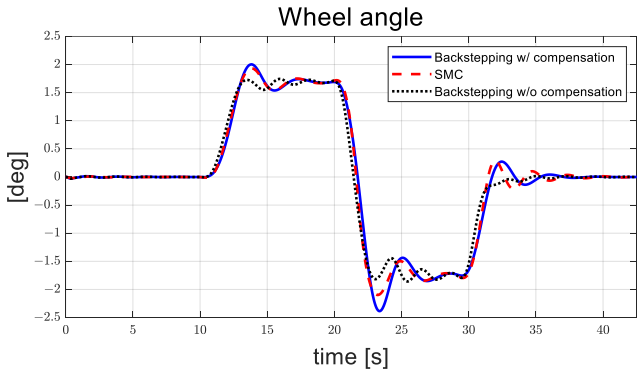


Figure 12. Scenario 1: vehicle wheel angles.

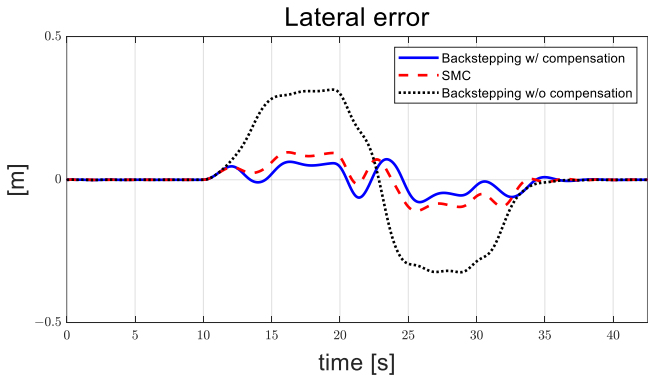


Figure 13. Scenario 1: lateral error.

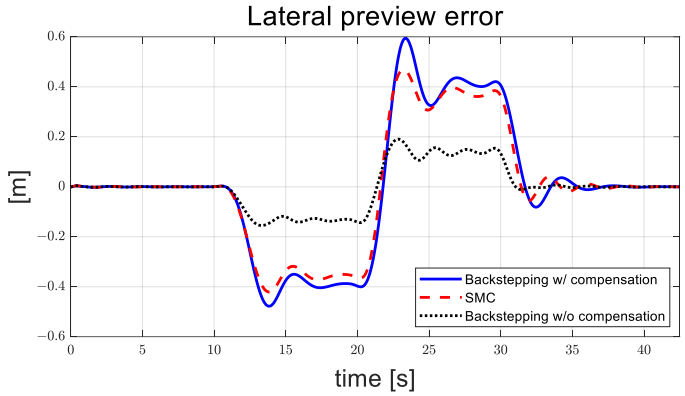


Figure 14. Scenario 1: lateral preview error.

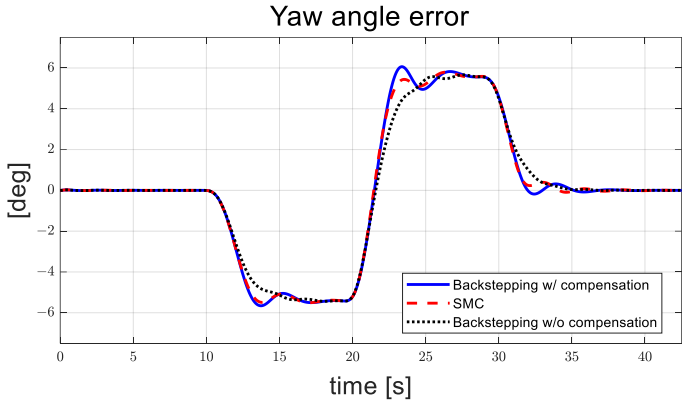


Figure 15. Scenario 1: yaw angle error (preview point).

Figure 9 shows the vehicle trajectory with the proposed algorithm, backstepping steering control without control delay compensation, and SMC. It can be confirmed that the proposed algorithm exhibits a lower lateral error within 0.08 m. However, the other controllers have a lateral error over 0.1 m. The longitudinal velocity was maintained at 30 km/h, which is consistent with the speed initially set for both controllers, as shown in Figure 10. The estimated control delay of the RLS is shown in Figure 11, with reasonable estimation performance after the control input is applied. However, it is difficult to secure estimation performance on a straight road, where less steering input is applied. Therefore, it is necessary to improve the control delay estimation algorithm in the future. The wheel angles are depicted in Figure 12 and show similar magnitudes to the control input. The control errors, namely the lateral error, lateral preview error, and yaw angle error at the preview point, are shown in Figures 13–15. The maximum lateral errors are 0.0782 m, 0.1092 m, and 0.3229 m. The maximum lateral preview errors are 0.5944 m, 0.4690 m, and 0.1908 m, respectively. In addition, the maximum yaw angle errors are 6.0655 deg, 5.7984 deg, and 5.6643 deg. Table 3 presents an analysis of the control errors for both controllers.

Table 3. Control error analysis.

Division		Backstepping w/ Compensation	SMC	Backstepping w/o Compensation
Lateral error	Max	0.0782 [m]	0.1092 [m]	0.3229 [m]
	Average	0.0226 [m]	0.0337 [m]	0.1197 [m]
	Standard deviation	0.0249 [m]	0.0379 [m]	0.1330 [m]

Table 3. *Cont.*

Division		Backstepping w/ Compensation	SMC	Backstepping w/o Compensation
Lateral preview error	Max	0.5944 [m]	0.4690 [m]	0.1908 [m]
	Average	0.1730 [m]	0.1542 [m]	0.0582 [m]
	Standard deviation	0.1944 [m]	0.1733 [m]	0.0664 [m]
Yaw angle error	Max	6.0655 [deg]	5.7984 [deg]	5.6643 [deg]
	Average	2.2209 [deg]	2.2005 [deg]	2.1331 [deg]
	Standard deviation	2.5093 [deg]	2.4772 [deg]	2.3868 [deg]

4.3. Evaluation 2—Scenario 2 (Four-Curved Loop Path)

As in Scenario 1, in Scenario 2, featuring four-curved loop path tracking, the three controllers were compared, and the results of the evaluation are shown in Figures 16–22.

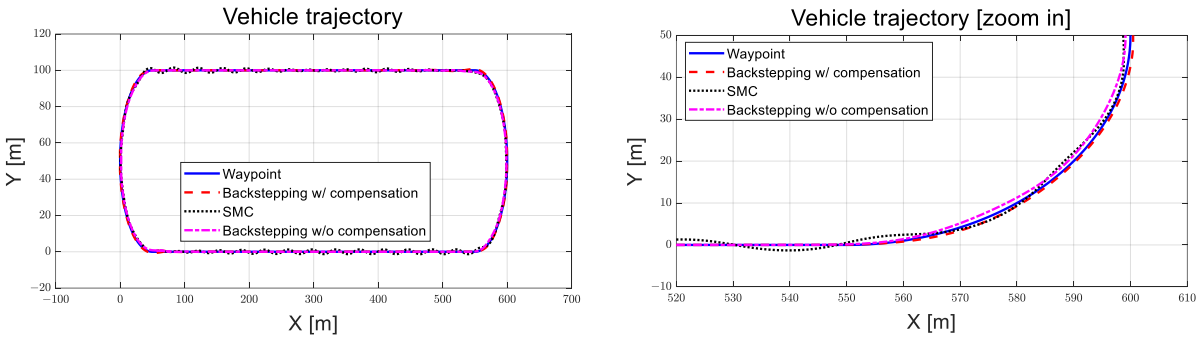


Figure 16. Scenario 2: vehicle trajectory.

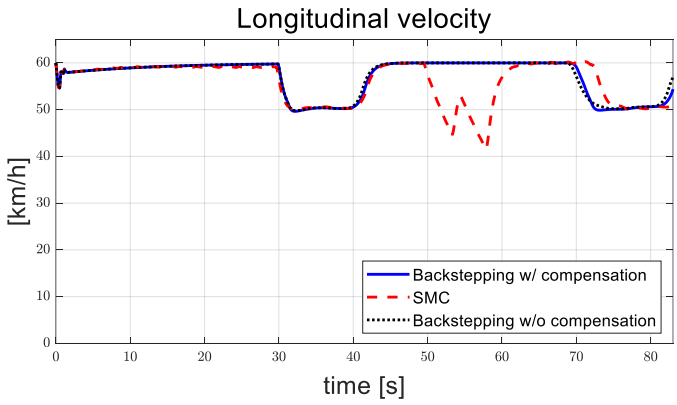


Figure 17. Scenario 2: vehicle longitudinal velocity.

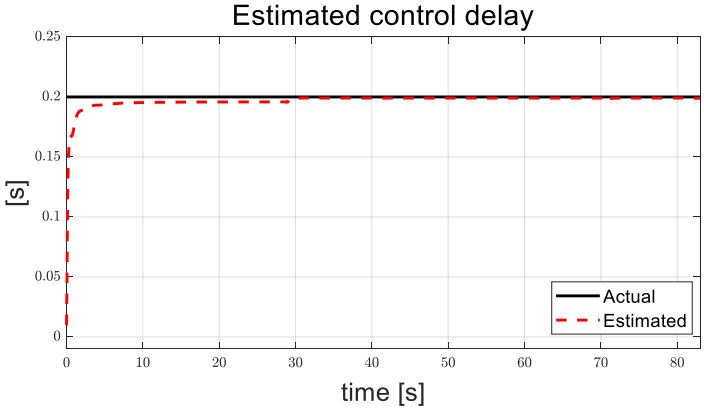


Figure 18. Scenario 2: estimated control delay of the steering system.

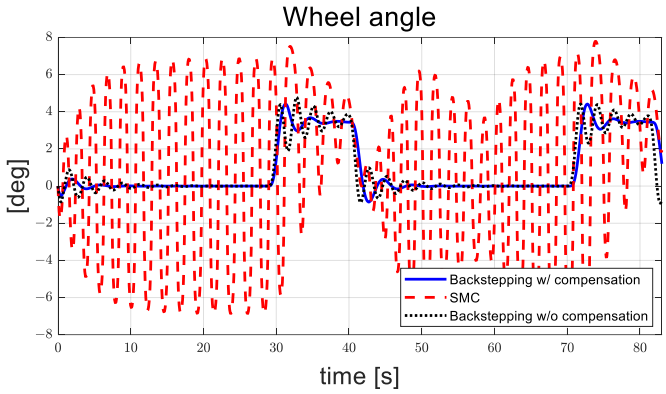


Figure 19. Scenario 2: vehicle wheel angles.

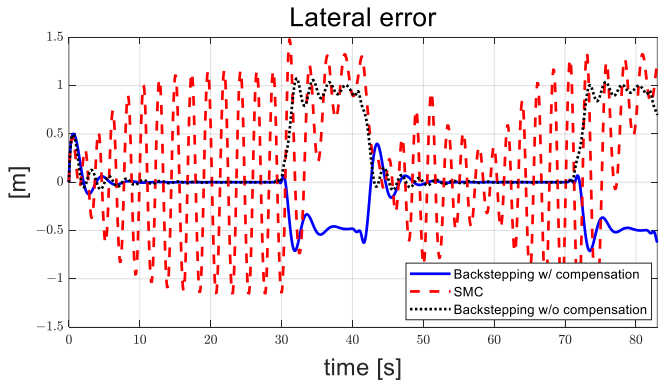


Figure 20. Scenario 2: lateral error.

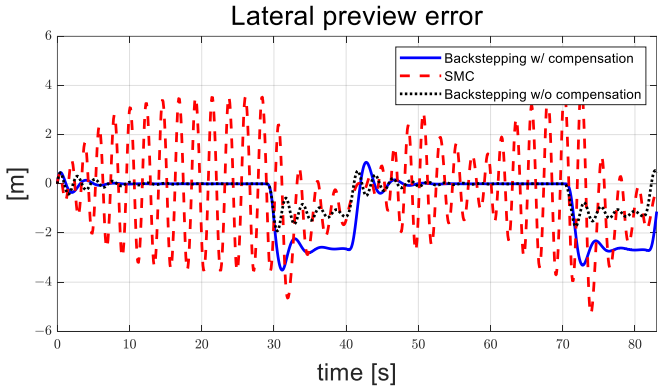


Figure 21. Scenario 2: lateral preview error.

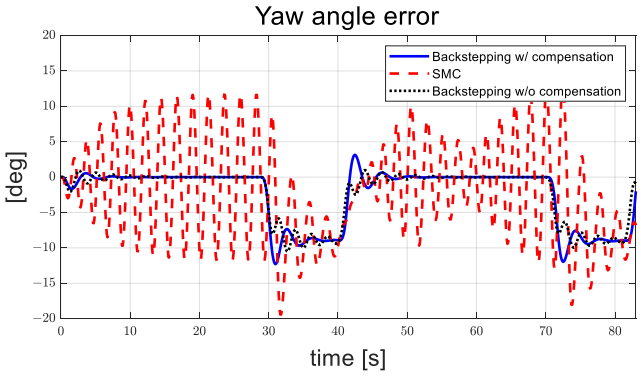


Figure 22. Scenario 2: yaw angle error (preview point).

Figure 16 shows the vehicle trajectory with the three controllers on a four-curved-loop path. Since the same parameters were applied, the SMC exhibits significant chattering phenomena due to improperly determined control parameters, whereas the proposed algorithm shows reasonable performance with a lateral error of about 0.8 m. The longitudinal velocity of the vehicle varied when the control input was applied under backstepping steering control. However, the longitudinal velocity was unstable at 50–60 s under SMC-based steering control because of the chattering phenomenon, as shown in Figure 17. As illustrated in Figure 18, the estimated control delay of the RLS demonstrated reasonable estimation performance after the control input was applied. Although the magnitudes of the control input and wheel angle of the backstepping control were similar, the SMC exhibited chattering phenomena on both straight and cornering roads, as shown in Figure 19. The control errors, including the lateral error, lateral preview error, and yaw angle error at the preview point, are compared and analyzed in Figures 20–22 and Table 4. The proposed steering control algorithm has lower control errors compared to the other controllers; however, the yaw angle error at the preview point is greater than that of the backstepping controller without compensation because the control input has a larger value than the backstepping controller without compensation. The SMC has the largest control errors, measuring 1.4813 m, 5.2869 m, and 19.4741 deg, respectively. It can be confirmed that the proposed algorithm has reasonable performance because the control errors have the lowest values.

Table 4. Control error analysis.

Division		Backstepping w/ Compensation	SMC	Backstepping w/o Compensation
Lateral error	Max	0.7120 [m]	1.4813 [m]	1.0748 [m]
	Average	0.1587 [m]	0.6293 [m]	0.2700 [m]
	Standard deviation	0.2229 [m]	0.3852 [m]	0.3972 [m]
Lateral preview error	Max	3.5123 [m]	5.2869 [m]	1.9212 [m]
	Average	0.7725 [m]	1.6125 [m]	0.3716 [m]
	Standard deviation	1.1605 [m]	1.1369 [m]	0.5230 [m]
Yaw angle error	Max	12.2975 [deg]	19.4741 [deg]	10.5116 [deg]
	Average	2.6112 [deg]	6.2881 [deg]	2.4428 [deg]
	Standard deviation	3.9402 [deg]	4.0023 [deg]	3.6537 [deg]

5. Conclusions

In this study, a Lyapunov approach-based backstepping control has been proposed, considering the estimated control delay in real time. Control delays, such as time delay and actuator lag in the steering system of autonomous mobility, can affect control performance, causing performance and control stability degradation and also compromising operator safety. To address the control delay phenomenon, a backstepping steering control based on the Lyapunov approach has been proposed. The RLS was used to calculate the control delay in real time. The performance of the proposed control algorithm was evaluated using MATLAB/Simulink and CarMaker software under S-curved and four-curved loop path scenarios by comparing SMC, and the backstepping controller without compensation. The reasonableness of the proposed algorithm was validated by achieving the lowest control errors in both scenarios. However, it was necessary to advance the control delay estimation algorithm to ensure better estimation performance on straight roads. In addition, the validation was conducted in a simulation environment, and the uncertainties and disturbances were not considered. Therefore, an actual test platform-based outdoor evaluation is planned for future work. We hope that our method will be helpful not only in the mobile robotics industry, especially in the area of robust actuator control considering control and communication delays in over-actuated systems, such as over-actuated snake robots [53], but also in research focused on resilient robots [54].

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